



Measurements and simulations of ignition delay times and laminar flame speeds of nonane isomers

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ABSTRACT

Ignition delay times (IDTs) and laminar flame speeds (S_L) of C_9H_{20} (nonane) isomers are systematically investigated. IDTs of normal nonane ($n-C_9$), 2-methyloctane ($2mC_8$), 2,4-dimethylheptane ($24mC_7$), and 2,2,4,4-tetramethylpentane ($2244mC_5$) are experimentally obtained by a shock tube facility and numerically simulated by a chemkin 0-D reactor model. Further, laminar flame speeds (S_L) of $n-C_9$ and $2244mC_5$ are measured by spherical expanding flames in a constant-temperature, constant-pressure dual-chamber cruciform burner over a wide range of the equivalence ratio ($\Phi = 0.7$ – 1.4), which are used to compare with numerically simulated results obtained by chemkin 1-D flame speed model. Detailed reaction mechanisms of KUCRS, LLNL and JetSurF ver.02 are used for numerical simulations. It is found that experimental IDTs increase with the number of methyl branches, especially in low-temperature and negative temperature coefficient (NTC) regions, where the increase of IDT with the number of methyl branches are well predicted by KUCRS. We also find that the measured values of S_L of highly branched $2244mC_5$ are smaller than those of $n-C_9$ at all values of Φ studied, of which measured S_L data are successfully reproduced by the 1-D flame speed model with KUCRS. These results are important to our understanding of reaction characteristics for highly branched nonane isomers and for the designing of optimal alternative fuels in internal combustion engines.

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1. Introduction

Due to strong demands for prediction and control of fossil fuel combustion, chemical reaction characteristics of hydrocarbons (HCs) have been extensively studied. As a result, fundamental knowledge on the reaction characteristics of fossil-fuel relevant HCs has been obtained [1–13]; various chemical reaction mechanisms have been proposed to drastically improve predictions and control of practical combustors [14–16].

However, more detailed knowledge is still required for the designing of optimal alternative fuels to substantially reduce CO_2 emission from combustion. For example, next generation alternative fuels currently proposed are composed of highly or weakly branched saturated hydrocarbons [17,18]. Although the reaction

characteristics of isomers of saturated HCs have been investigated (e.g., C_6 [19], C_7 [20–22], C_8 [22–24] and $C_7 \sim C_{20}$ [25]), fundamental knowledge is still needed to improve the modeling and predictions of the combustion of those HCs. As such, the objective of this study is to integrate the fundamental knowledge on the reaction characteristics of hydrocarbon fuels especially focusing on C_9 isomers. In our previous research [26], ignition delay times (IDTs) of C_9H_{20} isomers, i.e. n -nonane ($n-C_9$) and 2,2,4,4-tetramethylpentane ($2244C_5$) were measured with a newly developed shock tube facility to clarify the combustion characteristics of highly branched HCs. Although the ignition delay time of normal nonane has been reported by several researchers [27–30], the work in [26] was the first investigation of the highly branched nonane isomer.

In this study, we investigate the reaction characteristics of C_9H_{20} isomers in more depth via comprehensive comparisons between experimental measurements and numerical simulations. In addition to $n-C_9$ and $2244mC_5$, target species are also extended to mono- and di-methyl isomers such as 2-methyloctane ($2mC_8$) and

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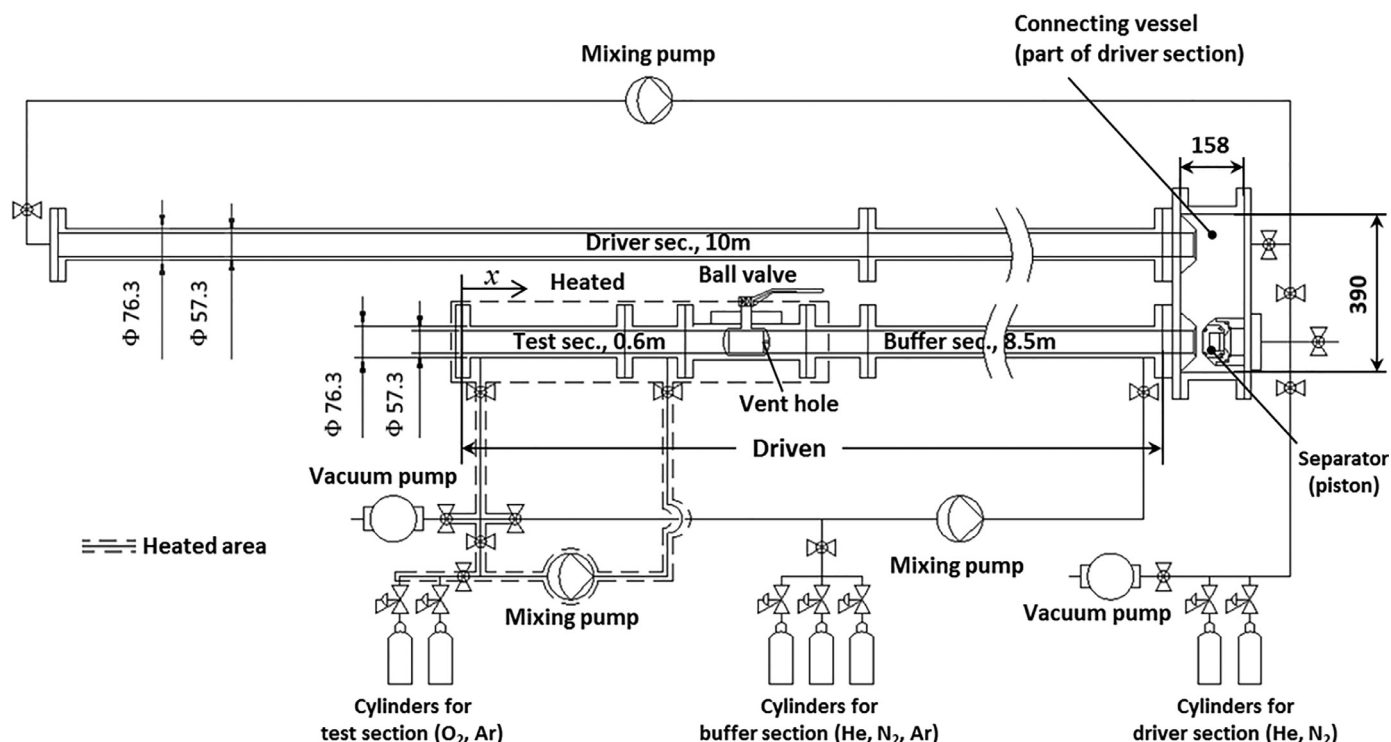


Fig. 1. Schematic of shock tube facility [26].

2,4-dimethylheptane (24mC₇). To assess the validity of simulated combustion characteristics, laminar burning velocities (S_L) of n-C₉ and 2244mC₅ are also measured. S_L is the speed of an unstretched laminar premixed flame with respect to unburned reactants, it is the fundamental physical-chemical property of a flammable mixture, and it is the key input parameter of various models for premixed turbulent combustion to simulate turbulent burning in internal combustion engines. To the best knowledge of the authors, data of S_L for nonane isomers, such as 2244mC₅, are still not available. Hence, we measure values of S_L for n-C₉ and 2244mC₅ as a function of Φ using laminar spherical expanding flames in a fan-stirred explosion bomb, so that comparisons with those predicted by the present numerical simulations can be made and the integration of the fundamental knowledge on the reaction characteristics of C₉ isomers can be obtained.

2. Experimental methods

2.1. Ignition delay time measurements

Fig. 1 shows the schematic of the shock tube used in this study. The detailed system geometries and validation have been described in [26]. For completeness, a brief description is included. The shock tube system consists of a driver section and a driven section, where the latter section is further divided into buffer and test sections, that is, a three sectional configuration is adopted as a “Constraint Volume Strategy” proposed by [31]. As illustrated in Fig. 1, the driver section is composed of driver pipes and a rectangular “connecting vessel”. The driver pipes are 57.3 mm in inner diameter with a total length of 10 m, and the rectangular connecting vessel is 158 mm wide, 390 mm long and 140 mm high. In the connecting vessel, a movable piston is installed to separate the driver and driven sections. The piston movement initiates the shock wave that propagates into the driven (buffer) section. As a result of the thorough validation by our previous work [26], we

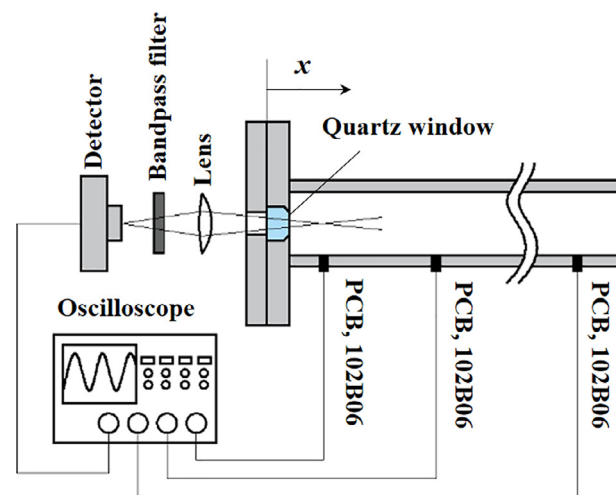


Fig. 2. Measurement system of pressure and OH* emission in the test section [26].

confirm that the vented geometry has little effect on the shock waves in the experimental range of this research.

On the lower pressure side, the driven section is a cylindrical tube with an inner diameter of 57.3 mm, which again are divided into both buffer and test sections. The lengths of buffer and test sections are respectively 8.5 m and 0.6 m (see Fig. 1) which are separated by closing a heat-resistant ball valve (TOYO KOATSU Co., Ltd.) installed between them. During the experiment, the test section and the ball valve are heated electrically to 150 °C to avoid fuel condensation.

The ignition delay times are determined based on the pressure variations and OH* emission. Fig. 2 shows the measurement system at the end of test section. Based on the signals of two pressure sensors installed at $x = 50$ mm and 150 mm upstream of the end plate, the velocity of incident shock wave as well as the ar-

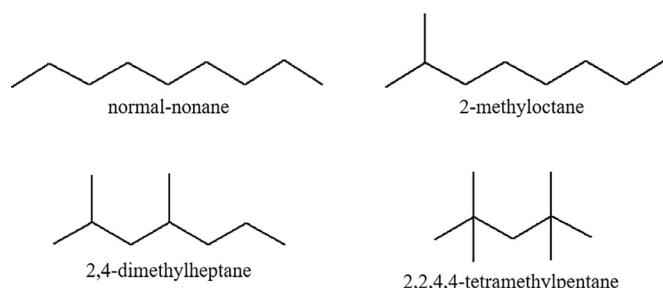


Fig. 3. Fuels tested in this study.

rival time of the incident shock wave to the end of the test section ($t = 0$ s) can be measured. We estimate the errors in manufacturing (less than ± 0.2 mm) and thermal expansion (less than 0.5 mm), which result in less than ± 7 K for temperature measurement and less than $1 \mu\text{s}$ uncertainty for the arrival time of incident shock wave ($t = 0$ s).

The onset of auto ignition is determined by the chemiluminescence of OH^* which is obtained with a photo detector (THORLABS, PDA#&A-EC) through an optically accessible quartz viewport installed at the end of the driven tube (Fig. 2, quartz window) and a bandpass filter centered at 306.0 nm. Note that the so-called pressure bump has been observed for several cases before the onset of auto-ignition. The pressure variation by the bump is less than 10% for all experimental runs in this study.

The IDT measurements are conducted by using the same procedure as in [26]. Before a run, all sections are evacuated without separations. The driver and driven sections are then separated by closing the movable piston. After that, the driver section is filled with He and N_2 ; the driven section is supplied with Ar, N_2 and He. Those gases are mixed by the mixing pumps installed in each of both sections for 10 min. Then, the ball valve is closed to separate the buffer and test sections. The test section is evacuated and liquid fuel (C_9H_{20} isomer) is supplied into the test section by a micro-syringe. Thereafter, Ar and O_2 are supplied to the test section and are mixed using the mixing pump. Finally, by opening the separation ball valve carefully, a run begins by releasing the piston between the driver and driven sections to initiate the shock wave that propagates in the driven section.

In this study, IDTs of C_9H_{20} isomers are examined, which are respectively n-nonane (98.0% purity), 2-methyloctane (98.0% purity), 2,4-dimethylheptane (98.0% purity), and 2,2,4,4-tetramethylpentane (99.0% purity), as indicated in Fig. 3. Hereafter, these four fuels are respectively denoted as “n-C₉”, “2mC₈”, “24mC₇” and “2244mC₅”. As for the oxidizer, the O_2 volume fraction is fixed at 21% considering the practical combustors run with air. It is worth noting that the ideal constant pressure (CP) condition cannot be achieved in this study, although the same “three-sectional configuration” as the CRV (Constrained Reaction Volume) strategy is adopted. This is because the amount of fuel, and hence, heat release of the test mixture is not suppressed to fix the oxygen concentration at 21%. The benefits of the three-sectional configuration in this study are: (1) The quantity of fuel used in each experimental run is reduced and (2) the heating area to inhibit the fuel condensation is limited only to the test section. As for the latter benefit, we confirm that the temperature of all inner surfaces contacting the test gas is heated uniformly to 120 °C within the deviation of ± 5 °C. A well-defined test mixture can be thus prepared via the temperature-controlled test section [26].

For equivalence ratio conditions of 0.5, 0.8, 1.0 and 1.5, the ignition delay times at 1.0 MPa are examined for the temperature range varying from 630 K to 1418 K ($1000/T$: 0.705–1.59). The gaseous composition and pressure in each section are determined

so that the shock impedance when the incident shock wave passes through each section satisfies the tailored conditions. The gaseous thermodynamic properties to determine or estimate the experimental conditions are estimated by the method developed by Ritter and Bozzelli [32] and implemented in KUCRS (Knowledge-basing Utilities for Complex Reaction Systems) [33]. In the experiment, compositions (fractions) of gaseous species are determined by monitoring their partial pressure. Based on the resolution of pressure gauge, experimental errors caused by the deviation of gaseous composition are theoretically estimated to be less than $\Delta T = \pm 10$ K for all experimental conditions of this work. As for the pressure, the so-called pressure bump has been observed for several cases before the onset of auto-ignition. The pressure variation by the bump is less than 10% for all experimental runs in this study.

2.2. Measurements of laminar flame speeds

Expansion of n-C₉/air and 2244mC₅/air laminar spherical flames are experimentally investigated at 1 atm and 373 K with six different values of Φ (= 0.7, 0.9, 1.0, 1.1, 1.2, 1.4) using the constant-pressure, constant-temperature, dual-chamber cruciform burner to extract their corresponding unstretched laminar flame speeds, S_u^0 or S_L , with respect to unburned reactants. Since the cruciform burner and diagnostic techniques have been discussed in detail elsewhere [34–39], here only a brief description is included for completeness alongside a detailed description of the new modifications of the fuel pre-vaporization heating system for nonane isomers.

A simplified sketch of the inner large 3D cruciform burner is presented in the right part of Fig. 4. The inner 3D cruciform burner resides in a huge outer pressure vessel (not shown) and consists of a large cylindrical steel pipe placed horizontally with four smaller cylindrical steel pipes perpendicularly aligned and symmetrically welded around the central part of the inner burner. The intersecting central volume of the 3D cruciform burner is nearly spherical having a minimum diameter of about 300 mm. Two identical counter-rotating fans driven by two 10 HP motors together with two identical heated perforated plates are used to mix fuel and air before ignition using a pair of cantilevered pin-to-pin spark electrodes (see the inserted Schlieren image in Fig. 4). There are four large pressure-released valves symmetrically installed on the vertical cylindrical pipes of the cruciform burner, allowing a spherical laminar premixed flame to expand under a near-constant pressure condition. Twenty surface heaters cover the cruciform burner, and together with the two perforated plate heaters, they are capable of generating a central uniform constant high-temperature region of $100 \times 100 \times 100$ mm with only 1 °C variation for spherical flame speed measurements (please see Ref. [38] for a detailed treatment).

The left part of Fig. 4 is the fuel pre-vaporization system using surface heaters, where the heated fuel tank is vacuumed before the injection of sufficient liquid fuel from an electrically-controlled atomizer. Since the boiling point temperatures of nC₉ and 2244mC₅ are respectively 151 °C and 123 °C, we set the inside temperature of the heated fuel tank at 160 °C, measured by a thermocouple, to ensure the complete vaporization of liquid nC₉ and 2244mC₅ fuels.

The procedures to measure laminar flame speeds of nC₉/air and 2244mC₅/air mixtures at 373 K are as follows. First, all twenty surface heaters on the cruciform burner are turned on to a temperature of 160 °C for at least four hours prior to testing. Second, the two perforated plate heaters, which are carefully welded to a very long narrow serpentine heating strip [38]), are activated and set to a temperature of 250 °C. Third, the inner cruciform burner is vacuumed, the appropriate mole fraction of the pre-vaporized fuel from the heated fuel tank, either n-C₉ and 2244mC₅, is injected into the heated burner at a selected Φ by the partial pressure method, and

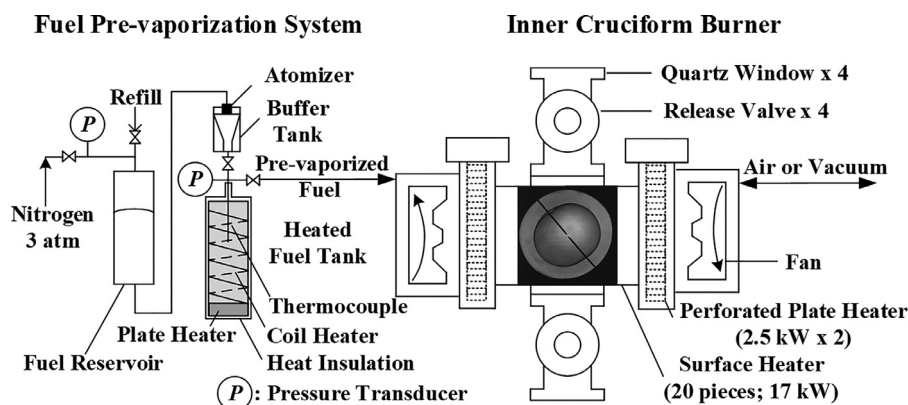


Fig. 4. Schematic of the cruciform burner for laminar flame speed measurements. Left: A liquid fuel pre-vaporization heating system. Right: The heated cruciform burner resided in a huge outer pressure vessel (not shown) [38].

then air is injected to fill up the burner to 1 atm. Great caution must be taken to carefully heat up all connected piping lines and joints between the fuel tank and the cruciform burner to a temperature of about 100 °C to make sure that an accurate mole fraction of the pre-vaporized $n\text{-C}_9$ can be obtained. Fourth, the two fans are counter-rotated at a fan frequency $f = 30$ Hz for four minutes to well mix the $n\text{-C}_9$ or 2244mC₅ mixture. It should be noted again that when the two counter-rotating fan-stirred streams pass through the two perforated plate heaters, an efficient turbulent heat convection is generated allowing a uniform temperature distribution in the experimentation domain. Fifth, after mixing, the fans are turned off and turbulence decays rapidly.

A run begins by centrally igniting the mixture by spark electrodes under a quiescent condition. We record the time evolutions of Schlieren images of centrally-ignited, outwardly-propagating spherical flames using a CMOS high-speed camera (Phantom 711) operated at 5000 frames/s with an 800×800 pixel resolution. Each Schlieren flame image is processed to obtain an area $A(t)$ enveloped by the flame and its radius $R_f(t) = \sqrt{A(t)/\pi}$. A typical example of the Schlieren flame image is given in Fig. 4 (the inserted image), where the mixture is $n\text{-C}_9/\text{air}$ at $\Phi = 0.7$, $R_f = 40$ mm, and $t = 15$ ms after ignition. Because the burned products are almost motionless in the expanding laminar spherical flame [40], the measured R_f corresponds to that of the downstream boundary of the flame. As such, the associated flame speed is equal to the instantaneous stretched flame speed $S_b = dR_f/dt$ with the stretch rate $\kappa = (1/A) dA/dt = (2/R_f) dR_f/dt$, where the subscript b represents the downstream burned state. Applying the most commonly-used linear relation, $S_b = dR_f/dt = S_b^0 - (2L_b/R_f) dR_f/dt$, with the assumption that the stretched flame speed data can be extrapolated to zero stretch where L_b is the Markstein length (please see [40] for detail), the downstream unstretched laminar flame speed, S_b^0 , can be evaluated for all values of Φ studied. Further, a nonlinear fitting method, $(S_b/S_b^0) \ln(S_b/S_b^0) = -2L_b/R_f$ [37,40], is also used for lean $n\text{-C}_9$ and 2244mC₅ only in the case of $\Phi = 0.7$ to make a comparison with the data obtained by the linear relation. Thus, the upstream unstretched laminar flame speed, $S_u^0 = S_L = (\rho_b/\rho_u)S_b^0$ can be determined, where ρ is the density and the subscript u refers to the upstream unburned state.

3. Numerical simulation

3.1. Numerical model

The ignition delay time is obtained numerically with the Chemkin-Pro 0-D homogeneous reactor model, while the laminar

flame speed is gained with the 1-D flame speed model [41]. The ignition delay times are calculated with a constant volume (CV) condition, and these numerical results are compared with the experimental results.

3.2. Kinetic models

As for the detailed reaction mechanisms, we apply the mechanisms for 2-methyl alkanes and n -alkanes (hereafter, “LLNL”) [42], JetSurF v2.0 [43] (“JetSurF”), and the mechanisms prepared with the auto generation program “KUCRS” (Knowledge-basing Utilities for Complex Reaction Systems) [33]. With the LLNL mechanism, both ignition delay times and laminar flame speeds for n -alkanes up to C16 and 2-methylalkane up to C20 can be simulated in wide ranges of pressure, temperature and equivalence ratio. In this study, IDT and S_L of $n\text{-C}_9\text{H}_{20}$ and 2mC₈ are obtained with LLNL. With JetSurF, both IDT and S_L can be simulated up to C12 as for n -alkanes, however, it includes only the high temperature chemistry. Then, in this study, IDT for $T > 1000$ K as well as S_L of $n\text{-C}_9$ are simulated with JetSurF. On the other hand, KUCRS is mainly developed to simulate the ignition delay times not only alkanes but also other various hydrocarbons, such as cycloalkane, alkene, alcohol, and so on. Thus, KUCRS can reproduce the ignition delay times of saturated hydrocarbons well, but it has not been validated for a laminar flame speeds. In fact, results of our preliminary numerical simulation on S_L showed that KUCRS did not predict S_L of $n\text{-C}_9$ and 2244mC₅ well. This is because that the original KUCRS omits several intermediate C2–C4 hydrocarbon species (e.g., C_3H_4 , $\text{C}_3\text{H}_5\text{O}$, etc.) and their associated reactions. Therefore, this study modifies the reaction mechanisms generated by KUCRS based on the gasoline surrogate mechanisms [15] and the AramcoMech 2.0 [44–50] for a better prediction of S_L (see Fig. S1-1 in the Supplemental Material 1). Note that such modification has little effects on IDT in the target range of this research (see Fig. S1-2 in the Supplemental Material 1). In short, in this study, IDT and S_L of all C_9H_{20} isomers, i.e. $n\text{-C}_9$, 2mC₈, 24mC₇ and 2244mC₅, are simulated by the modified KUCRS mechanisms that include: (1) 669 species and 1784 reactions for $n\text{-C}_9$, (2) 1258 species and 3314 reactions for 2mC₈, (3) 1777 species and 4497 reactions for 24mC₇, (4) 425 species and 1190 reactions for 2244mC₅, (5) 1009 species and 2658 reactions for $n\text{-C}_9$ / 2244mC₅ mixture, and (6) 1461 species and 3898 reactions for 2mC₈ / 2244mC₅ mixture. These KUCRS-based detailed reaction mechanisms and associated thermal and transport databases are provided in the Supplemental Material alongside the lists of species (name and structure).

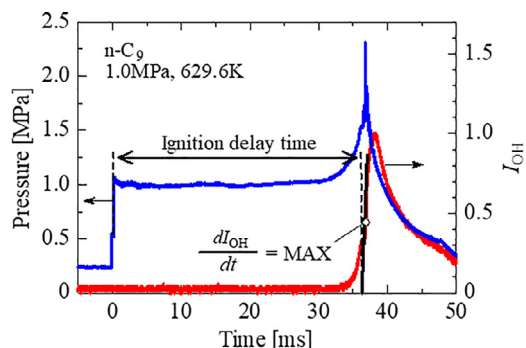


Fig. 5. Pressure and OH* history of a typical test for $n\text{-C}_9/\text{Ar}/\text{O}_2$ at $P = 1.0$ MPa, $\Phi = 1.0$ and $T = 629.6$ K.

4. Results

4.1. IDT of C_9 isomers

4.1.1. IDT of $n\text{-C}_9$

First, IDT for $n\text{-C}_9$ is examined. Fig. 5 shows a typical example of temporal variations of pressure and OH* emission obtained for $n\text{-C}_9$ at $\Phi = 1.0$, $P = 1.0$ MPa and $T = 630$ K. Pressure is measured by a pressure sensor installed 50-mm upstream of the test section closed end, and $t = 0$ is the arrival time of the incident shock wave at the sensor. As shown in Fig. 5, pressure increases promptly at $t = 0$, remains constant at the target pressure of 1.0 MPa for more than 30 ms, and then steeply increases at $t = 36.7$ ms due to auto ignition. Such flat pressure profile is considered to be mainly attributed to the CRV technique which can avoid remote ignition. In this study, IDT is determined by using the temporal variation of OH* luminosity at the period of time from the arrival of the reflected shockwave ($t = 0$) to the onset of auto ignition ($t = 36.7$ ms) where the time differentiation of the OH* intensity in an arbitrary unit reaches its maximum, $dI_{\text{OH}}/dt = \text{MAX}$, as indicated in Fig. 5. Note that IDTs in numerical simulations are also estimated by the same manner using numerically obtained OH radical fraction.

In Fig. 6, numerical IDTs of $n\text{-C}_9$ obtained by KUCRS (Fig. 6a), LLNL (Fig. 6b), and JetSurF (Fig. 6c) for various equivalence ratios ($\Phi = 0.5, 0.8, 1.0$ and 1.5) are shown together with experimental results for comparison. Again, the numerical result with JetSurF is shown only for $1000/T < 1.0$. Pressure is fixed at 1.0 MPa. Here, experimental results show the IDTs obtained in each experimental run without averaging multiple runs. In Fig. 6(a), it can be seen that the numerical results with KUCRS underestimate the experimental IDT in the high temperature region ($0.8 < 1000/T <$

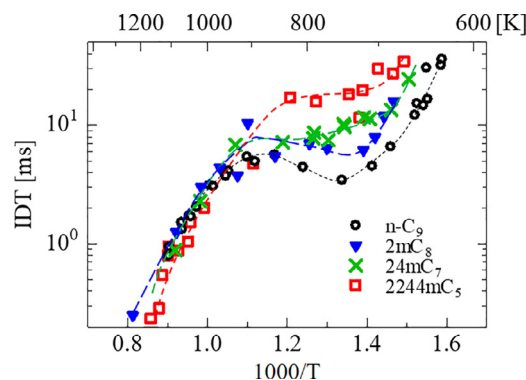


Fig. 7. Ignition delay times of C_9H_{20} isomers ($P = 1.0$ MPa, $\Phi = 1.0$).

1.0) for all conditions of Φ , while show better agreement in middle to low temperature region ($1000/T > 1.0$). On the other hand, the numerical result with LLNL (Fig. 6b) shows good agreement in high ($1000/T < 1.0$) and low ($1000/T > 1.4$) temperature regions, whereas poor agreement is visible in the middle temperature region ($1.0 < 1000/T < 1.4$). JetSurF shows good agreement in the high temperature region ($1000/T < 1.0$), where the mechanism is well validated though the effect of Φ has little influence in this region.

4.1.2. IDT of C_9H_{20} isomers

Fig. 7 shows experimentally obtained IDTs of 2-methyloctane (2mC₈), 2,4-dimethylheptane (24mC₇) and 2,2,4,4-tetramethylpentane (2244mC₅) together with $n\text{-C}_9$ under the condition of $P = 1.0$ MPa and $\Phi = 1.0$. Note that the lines in Fig. 7 are drawn to guide the readers, which are not numerical data. In Fig. 7, it can be seen that the difference of IDT is prominent among various isomers in the low to middle temperature region ($1000/T > 1.1$), whereas it is small in the high temperature region ($1000/T < 1.1$). In the low to middle temperature region, IDT is the shortest for $n\text{-C}_9$ and it becomes longer with the increasing number of branches, in the order of $n\text{-C}_9 < 2\text{mC}_8 < 24\text{mC}_7 < 2244\text{mC}_5$. For example, at fixed temperature condition of $1000/T = 1.35 > 1.1$, IDT is 3.45 ms for $n\text{-C}_9$, 6.21 ms for 2mC₈, 10.24 ms for 24mC₇ and 30.0 ms for 2244mC₅. Unlike the large difference in the low to middle temperature region, IDT is nearly the same in the high temperature region ($1000/T < 1.1$) for all isomers studied.

Effects of equivalence ratio on the IDT are also examined for 2mC₈, 24mC₇ and 2244mC₅. In Fig. 8, experimentally obtained IDTs for 2mC₈ under various Φ conditions are shown and compared with numerical results with KUCRS (Fig. 8a) and LLNL (Fig. 8b). Note that the experimental IDT of 2mC₈ for $\Phi = 0.5$ can be ob-

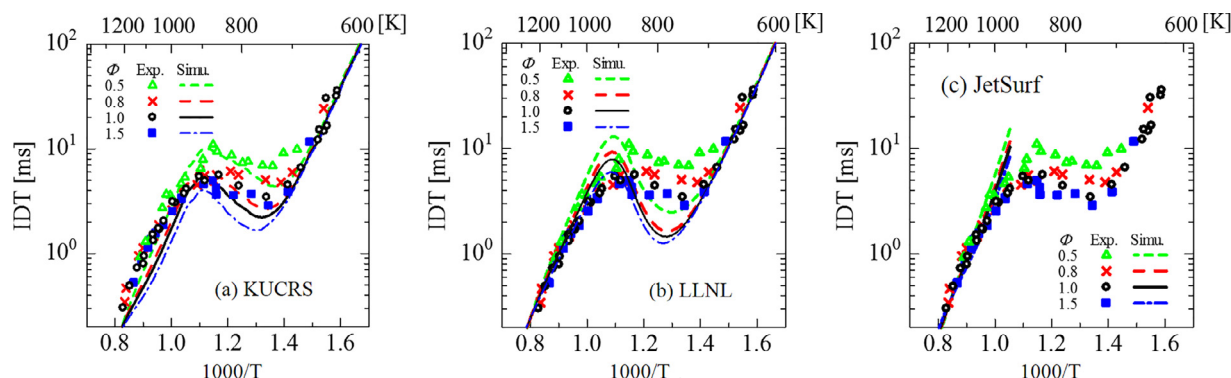


Fig. 6. Ignition delay time of $n\text{-C}_9/\text{Ar}/\text{O}_2$ for $\Phi = 0.8, 1.0$ and 1.5 with $P = 1.0$ MPa (symbols: experimental data, lines: numerical results with (a) KUCRS, (b) LLNL, and (c) JetSurF).

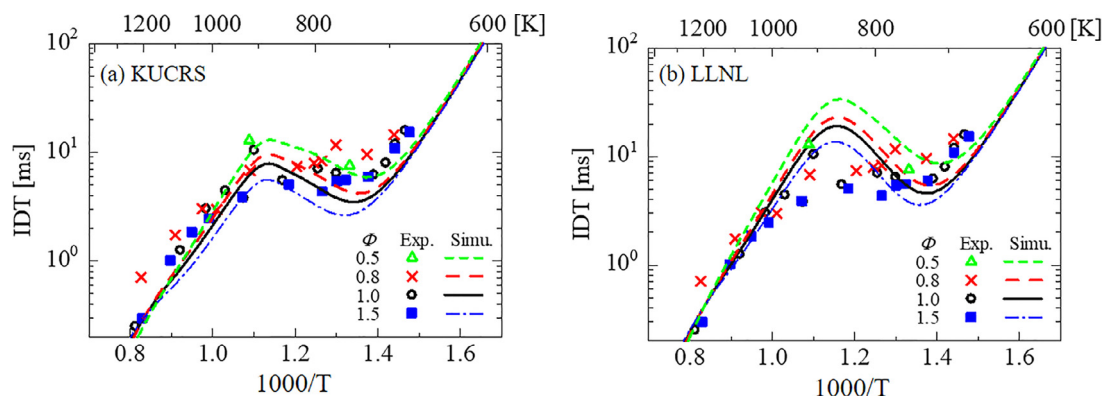


Fig. 8. Ignition delay time of 2mC₈/Ar/O₂ for $\Phi = 0.5, 0.8, 1.0$ and 1.5 with $P = 1.0$ MPa (symbols: experimental data, lines: numerical results with (a) KUCRS and (b) LLNL).

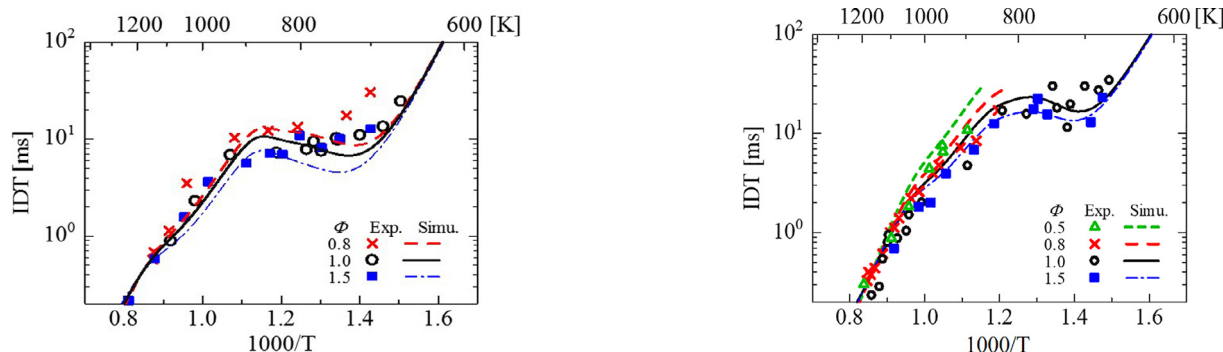


Fig. 9. Ignition delay time of 24mC₇/Ar/O₂ for $\Phi = 0.5, 0.8, 1.0$ and 1.5 with $P = 1.0$ MPa (symbols: experimental data, lines: numerical results with KUCRS).

tained under rather limited conditions, and numerical results with JetSurF are not shown because 2mC₈ is not included in JetSurF. As compared with experimental IDT data, the numerical results with KUCRS (Fig. 8a) deviate in the high ($1000/T < 1.0$) and slightly in low ($1000/T > 1.3$) temperature regions, but they agree well in the middle temperature region ($1.0 < 1000/T < 1.3$). On the other hand, as shown in Fig. 8(b), the numerical results with LLNL seem to show excess NTC tendency than the experimental results, that is, while the numerical results show good agreement in the high temperature region ($1000/T < 1.0$), however, significant overestimation in the range of $1.0 < 1000/T < 1.2$ and slight underestimation in the low temperature region of $1000/T > 1.3$ are observed.

In Fig. 9, experimental IDTs of 24mC₇ are shown together with numerical results with KUCRS. Note that IDT for $\Phi = 0.5$ cannot be obtained because mixture is hardly ignited. Comparing with the stoichiometric condition, experimentally obtained IDTs in the fuel rich condition ($\Phi = 1.5$) show nearly the same tendency in high ($1000/T < 1.1$) and low ($1000/T > 1.3$) temperature regions, while slightly shorter in the middle temperature region ($1.1 < 1000/T < 1.3$). For fuel lean conditions, IDT seems to increase especially in the low temperature region ($1000/T > 1.3$). The results of numerical simulation using KUCRS show reasonable agreement with the experimental results from high to middle temperature regions ($1000/T < 1.2$).

In Fig. 10, experimental IDTs of 2244mC₅ for various equivalence ratio conditions are shown together with numerical results with KUCRS. Note that IDT for $\Phi = 0.5$ can be obtained only in the range $1000/T > 1.1$ because the IDT is too long to be measured with our shock tube facility. IDT in the fuel rich condition ($\Phi = 1.5$) shows nearly the same tendency as the stoichiometric condition in almost all the experimental temperature range. For fuel lean conditions, IDT increases slightly with the decrease of Φ .

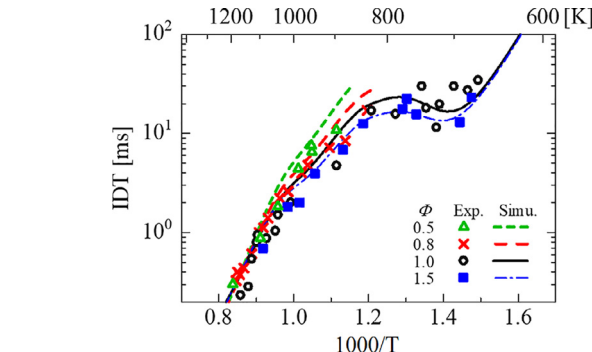


Fig. 10. Ignition delay time of 2244mC₅/Ar/O₂ for $\Phi = 0.5, 0.8, 1.0$ and 1.5 with 1.0 MPa (symbols: experimental data, lines: numerical results with KUCRS).

The results of numerical simulation with KUCRS show good agreement with the experimental results in the temperature range examined in this study.

4.1.3. IDT of binary mixture of C₉ isomers

IDTs of n-C₉, 2mC₈ and 2244mC₅ are obtained for various conditions, however, practical fuels are supposed to be a mixture of these isomers. It is well known that ignition delay times of hydrocarbon mixtures do not show linear tendency for composition [10]. In our previous work, it was found that IDT of binary mixtures of n-C₉ and 2244mC₅ show a strong non-linear tendency for the proportion of n-C₉ and 2244mC₅, especially in the NTC region. Then, in this study, IDTs of a binary mixture of 2mC₈ / 2244mC₅ are examined under $\Phi = 1.0$ and the constant temperature condition of 750 ± 15 K. Fig. 11 shows the results obtained by experiment and numerical simulation with KUCRS for various proportions of 2mC₈ and 2244mC₅. In Fig. 11, the vertical axis is IDT in linear scale and the horizontal axis is the volume fraction of 2mC₈ (V_{2mC8}); for example, “0%” in the horizontal axis corresponds to [2mC₈: 2244mC₅ = 0%: 100%], while “100%” is [2mC₈: 2244mC₅ = 100%: 0%]. IDTs of binary mixtures of n-C₉ / 2244mC₅ obtained in previous work [26] are also plotted in Fig. 11 against the volume fraction of n-C₉ (V_{n-C9}) indicated in the upper horizontal axis.

In Fig. 11, it can be seen that the IDT of mixed fuel shows strong non-linear dependency on the proportion of 2mC₈ and 2244mC₅. The IDT of 0% 2mC₈ (100% 2244mC₅) is 30 ms, however, an increase of the 2mC₈ fraction of just 10% results in the drastic decrease in IDT to 23 ms. An increase of the 2mC₈ proportion to 30% leads to an IDT of less than 10 ms, further increase in the 2mC₈ fraction leads to a gradual decrease in IDT and finally to 3.7 ms for 100% 2mC₈. Comparing IDT of 2mC₈ / 2244mC₅ mixture with that of n-C₉ / 2244mC₅ obtained in previous work [26], it confirmed

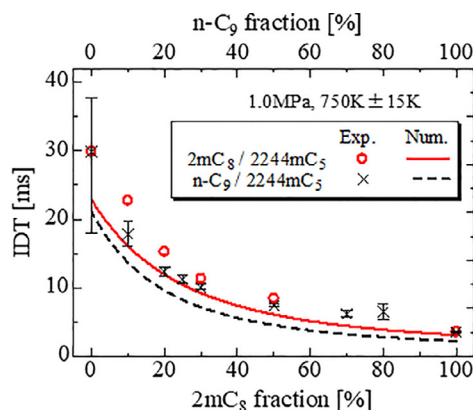


Fig. 11. Ignition delay time of binary mixture of 2mC₈ / 2244mC₅ and n-C₉ / 2244mC₅ obtained in previous work [26] (symbols: experimental data, lines: numerical results with KUCRS).

that IDT of 2mC₈ / 2244mC₅ mixture is in accordance with that of n-C₉ / 2244mC₅ mixture, which indicates that the highly branched isomer can strongly affect the IDT of mixed fuel only when its proportion is significantly higher than normal or weakly branched isomers.

As for the numerical results with KUCRS shown in Fig. 11, the experimental results are accurately reproduced in almost all proportion range despite the strong non-linearity of IDT. Hence, based on the aforesaid comparisons, it is suggested that KUCRS is an effective tool for predicting IDTs of the nonane structural isomers and their mixtures.

4.2. Laminar flame speeds of C₉H₂₀ isomers

In this study, the upstream unstretched laminar flame speeds (S_L) of C₉H₂₀ isomers are examined for two selected nonane isomers, i.e. n-C₉ and 2244mC₅, because they possess the largest difference in terms of the ignition delay time. Fig. 12(a) and (b) show typical sequential Schlieren images of centrally-ignited outwardly-propagating laminar spherical flame for n-C₉H₂₀/air and 2244mC₅/air mixtures under both lean ($\Phi = 0.7$) and rich ($\Phi = 1.4$) conditions with the effective Lewis number $Le \approx 2.9$ and 0.8 , respectively. Here Le is defined as the ratio of thermal diffusivity and mass diffusivity with the mass diffusivity being that of the deficient reactant and the abundant inert. The elapsed time after ignition is shown on the bottom left of each image in Fig. 12, where the images have a view field of 120×120 mm². Similar laminar spherical flame development is observed for both n-C₉ and 2244mC₅ cases for all equivalence ratio conditions examined in this research ($0.7 \leq \Phi \leq 1.4$), although the cellular structure can be observed on the flame surface for fuel rich conditions with $Le < 1$. Fig. 13 presents temporal variations of flame radii (R_f) and their corresponding downstream stretched flame speeds ($dR_f/dt = S_b$) using the central differentiation plotted against the stretch rate $\kappa = (2/R_f) dR_f/dt$, as typical examples, for n-C₉ (a,b) and 2244mC₅ (c,d), each fuel including three different $\Phi = 0.7, 1.0$, and 1.4 .

As shown in Fig. 13(b) for n-C₉ and Fig. 13(d) for 2244mC₅, the extrapolations to zero stretch for the determination of S_b^0 are performed only within linear, quasi-steady increase portions of the curves corresponding to $25 \text{ mm} < R_f < 45 \text{ mm}$ (30% of the burner radius). In the case of $\Phi = 0.7$ with $Le \approx 2.9 \gg 1$, both linear and non-linear fitting methods [34,37] are applied, where essentially the same values of S_b^0 are found. By using the density correction, the unstretched laminar flame speed on the unburned side, $S_L = (\rho_b/\rho_u) S_b^0$, can be thus measured.

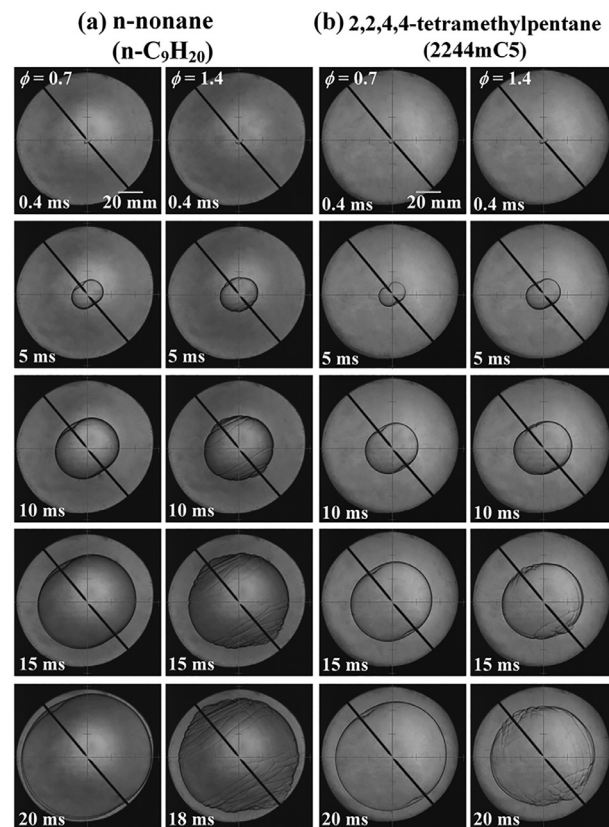


Fig. 12. High-speed Schlieren instantaneous images (120×120 mm²) taken at six different instants after ignition using a pair of pin-to-pin electrodes at a fixed inter-electrode gap $d_{\text{gap}} = 2$ mm: (a) n-C₉ and (b) 2244mC₅, each case having two different equivalence ratios $\Phi = 0.7$ and 1.4 .

Note that each S_L datum is obtained by the average of five repeated identical runs. For clarity, here we list values of S_L for n-C₉ / 2244mC₅ for comparison, each fuel having three different Φ : (1) $S_L = 35.27 / 32.78$ cm/s at $\Phi = 0.7$; (2) $S_L = 48.75 / 42.73$ cm/s at $\Phi = 1$; (3) $S_L = 37.99 / 27.71$ cm/s at $\Phi = 1.4$, respectively. At the $\Phi = 1.4$ case where $Le \approx 0.8 < 1$, the flame self-acceleration due to cellular structures can be clearly observed at larger R (the uprising curve of dR_f/dt in Fig. 13b and d). In short, it is found that all values of S_L for n-C₉ are greater than those of 2244mC₅ over the range of Φ varying from 0.7 to 1.4 .

Fig. 14(a) shows the measured laminar flame speeds of n-C₉ / air mixtures (symbols) as a function of Φ at $T = 373$ K, $P = 0.1$ MPa alongside three numerical results using KUCRS, LLNL, and JetSurF (lines) for comparison. It can be seen that experimentally obtained values of S_L reveal a typical tendency of S_L versus Φ for hydrocarbon / air mixtures, that is, S_L increases with Φ in the lean side and reaches its maximum value of 50.56 cm/s at $\Phi = 1.1$, and then, S_L decreases with Φ in the rich side. Note that we cannot find any previous S_L data of the n-C₉/air mixture at 373 K for a direct comparison, however, S_L data at higher temperatures of 403 K and 443 K were available [51,52]. Hence, we plot both present and previous n-C₉ S_L data as a function of Φ at three different temperatures for comparison (please see Fig. S2 in the Supplemental Material 2). Similar trend with the maximum S_L occurring at the same $\Phi = 1.1$ is found for all three temperatures; the higher the temperature, the higher S_L .

As for the numerical simulations, all three mechanisms can be used to predict the experimental S_L data as a function of Φ in a somewhat satisfactory degree. In particular, the results using KUCRS show an excellent agreement with experimental data over a

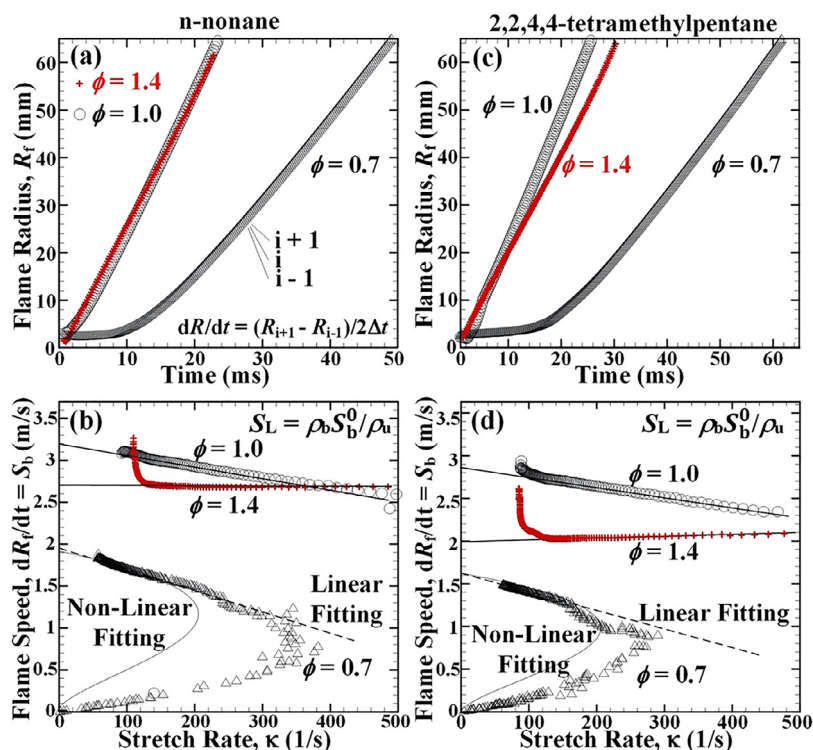


Fig. 13. Variations of flame radii with time (top row) and associated downstream flame speeds with the stretch rate (bottom row). Left column (a,b): n-C₉ and right column: 2244mC₅, each fuel including three equivalence ratios $\Phi = 0.7, 1.0$, and 1.4 . The superscript “0” of S_b^0 indicates the zero-stretch-rate state, while the subscripts “b” and “u” in the density ρ represent burned and unburned states.

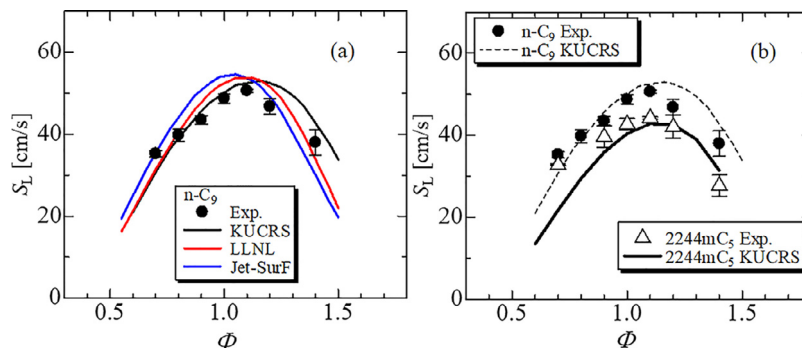


Fig. 14. Experimental and numerical laminar flame speeds of (a) n-C₉/air and (b) 2244mC₅/air mixtures at $T = 373$ K, $P = 0.1$ MPa.

range of $0.7 \leq \Phi \leq 1.1$, however, the KUCRS simulation slightly overestimates S_L in fuel rich conditions when $\Phi \geq 1.2$. In comparison, the JetSurF simulation overestimates S_L within $0.8 \leq \Phi \leq 1.1$, but it shows good agreement with experimental data at $\Phi = 1.2$ and 1.4 . Though the LLNL simulation slightly overestimates S_L around the stoichiometric condition ($0.9 \leq \Phi \leq 1.2$), it shows reasonably good agreement with experimental data for all values of Φ varying from $\Phi = 0.7$ to $\Phi = 1.4$.

Fig. 14(b) presents measured data of S_L vs. Φ for 2244mC₅ / air mixtures at $T = 373$ K, $P = 0.1$ MPa together with measured n-C₉ data and corresponding KUCRS simulations from Fig. 14(a) for comparison. It is found that at any given values of Φ , values of S_L for 2244mC₅ / air mixtures are lower than those of n-C₉ / air mixtures. Specifically, for 2244mC₅ / air mixtures at 100 °C and 1 atm, the maximum $S_L = 44.2$ cm/s occurs at $\Phi = 1.1$, whereas $S_L = 32.8$ cm/s at $\Phi = 0.7$ and $S_L = 27.7$ cm/s at $\Phi = 1.4$. These measured values of S_L as a function of Φ for the 2244mC₅/air mixture are new, which are important for numerical simulations and chemical kinetic predictions. It should be noted that, currently, only KUCRS can be used to simulate the measured S_L of

2244mC₅, as plotted as the solid line in Fig. 14(b), showing a very good agreement with the experimental data except for the data at $\Phi = 0.7$.

According to these results, it can be concluded that (1) IDT increases with the number of branches for C₉H₂₀ isomers and (2) values of S_L of highly branched isomers (2244mC₅) are lower than those of n-C₉ at any given Φ . As to the numerical simulations, the mechanism generated by KUCRS can be used to effectively predict both IDT and S_L of C₉H₂₀ structural isomers especially for the highly branched isomer of 2244mC₅. These findings should be useful to the development of surrogates for next generation biofuels.

5. Discussion

5.1. Difference of IDT in NTC and low-temperature region

Multiple complex chemical processes contribute to the difference of reactivity of isomers, especially in the low-temperature and NTC regions. The low temperature oxidation pathway of n-C₉ is illustrated in Fig. 15 as an example. In the case of n-C₉,

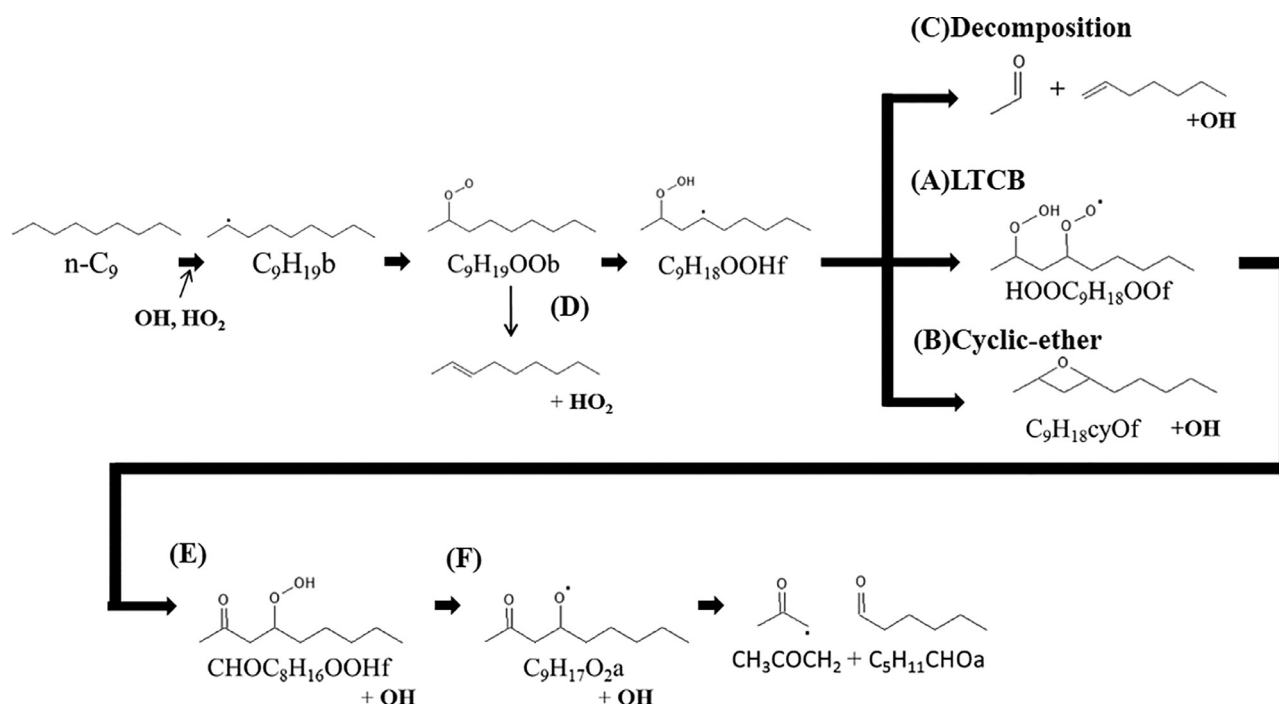


Fig. 15. Representative reaction low temperature oxidation reaction pathway for n-C₉.

starting from the H abstraction, the original fuel (n-C₉ in Fig. 15) is mainly converted to C₉H₁₉OO. Then, the reaction pathway is branched into two, i.e. (1) elimination of HO₂ from alkylperoxy radicals and (2) the formation of QOOH radicals. After that, the reaction path branches into three, i.e. (1) the reaction leading to low-temperature chain branching (LTCB) and inhibitive pathway, (2) the formation of cyclic ethers, and (3) the beta scission of QOOH radicals. Competitions of these reaction is considered to control the difference of reactivity of isomers.

On the reactivity difference of mono-methyl alkanes from normal alkane, Sarathy et al. [25] pointed out that LTCB becomes prohibitively slow for mono-methylated alkanes depending on the initial H-atom abstraction position, and instead, cyclic ether is rapidly formed. Wang et al. [18] investigated the reactivity of 2-methylheptane and 3-methylheptane, and they showed that 3-methylheptane is less reactive than 2-methylheptane in low-temperature and NTC regions. Further, Wang et al. [18] argued that the six-member ring isomerization which leads to low temperature branching is inhibited for 3-methylheptane depending on the initial H-atom abstraction position. Consequently, the cyclic ether formation reactions are promoted. Similar effect has also been argued for the di-methylated isomer (2,5-dimethyl hexane) by Sarathy et al. [23].

Considering the aforementioned results [18,23,25], this study performs reaction path analyses and brute force sensitivity analyses for nonane isomers, i.e. n-C₉, 2mC₈, 24mC₇, and 2244mC₅, focusing on reaction pathways leading to the low-temperature branching and the cyclic ether formation in the low temperature oxidation process. The sensitivity coefficient S for the aforesaid processes of (i) ~ (iv) on ignition delay times can be obtained by the following equation,

$$S = \frac{\log(\tau_{2k}/\tau_{0.5k})}{\log(2k/0.5k)} \quad (1)$$

where τ_{2k} is the ignition delay time calculated with the doubled rate coefficient and $\tau_{0.5k}$ is the ignition delay time calculated using the halved rate coefficient [30]. The analyses are made for the case of 1.0 MPa, 750 K and $\Phi = 1.0$ where the difference

of IDT between n-C₉ and 2244mC₅ is the maximum. Note that, for 2mC₈ (2-methyl octane), similar results as 2-methylheptane [25] are found. It is reasonable because the structural difference between 2-methylheptane and 2-methyloctane (the present study) is just the length of the straight chain, and no essential changes in the reaction pathway are expected. Then, detailed results of analyses for 24mC₇, 2244mC₅ are described in this paper.

Fig. 16 shows the results of sensitivity analysis for 24mC₇ and 2244mC₅, together with n-C₉. In Fig. 16, the negative sensitivities mean that the reaction shortens IDT when its rate is increased and vice versa. Analyses are also made for the other LTO (low temperature oxidation) related important reactions (D, E and F) in Fig. 15, as shown in Fig. 16. The sensitivity of each class of reactions is indicated by a color range bar. For example, 26 structural isomers exist for QOOH in the oxidation process of n-C₉, and each class of reaction for respective isomer shows different sensitivity on IDT. Then in Fig. 16, the sensitivity of individual reaction on IDT for different isomers is indicated by an open circle. For n-C₉ (Fig. 16a), the sensitivities of LTCB reactions are evidently negative whereas they are positive for the cyclic ether formation and decomposition. The reaction with the largest absolute value of LTCB is illustrated in Fig. 17(a) with reaction flux. It can be seen that the branching fraction to second O₂ addition (55.1%) exceeds that of the other two pathways, cyclic ether formation (39.7%) and decomposition (5.2%).

Fig. 17(b) and (c) show the representative reaction pathway of 24mC₇. Note that all initial reaction pathways to form QOOH from the original 24mC₇ are illustrated in Fig. S3 in the Supplemental Material 3. In Fig. 17(b), the low temperature oxidation reaction of 24mC₇ is initiated by the H-atom abstraction from secondary sites. As a result, branching fractions are similar to n-C₉ as presented in Fig. 17(a). On the other hand, as shown in Fig. 17(c), when the process is initiated by the hydrogen abstraction from the tertiary site, the second O₂ addition is hindered because of the lack of the six-member ring isomerization leading to ketohydroperoxide formation. Consequently, the branching fraction to the cyclic ether formation is significantly intensified (77.1%). Note that, cyclic ether formation reaction in Fig. 17(c) has negative sensitivity

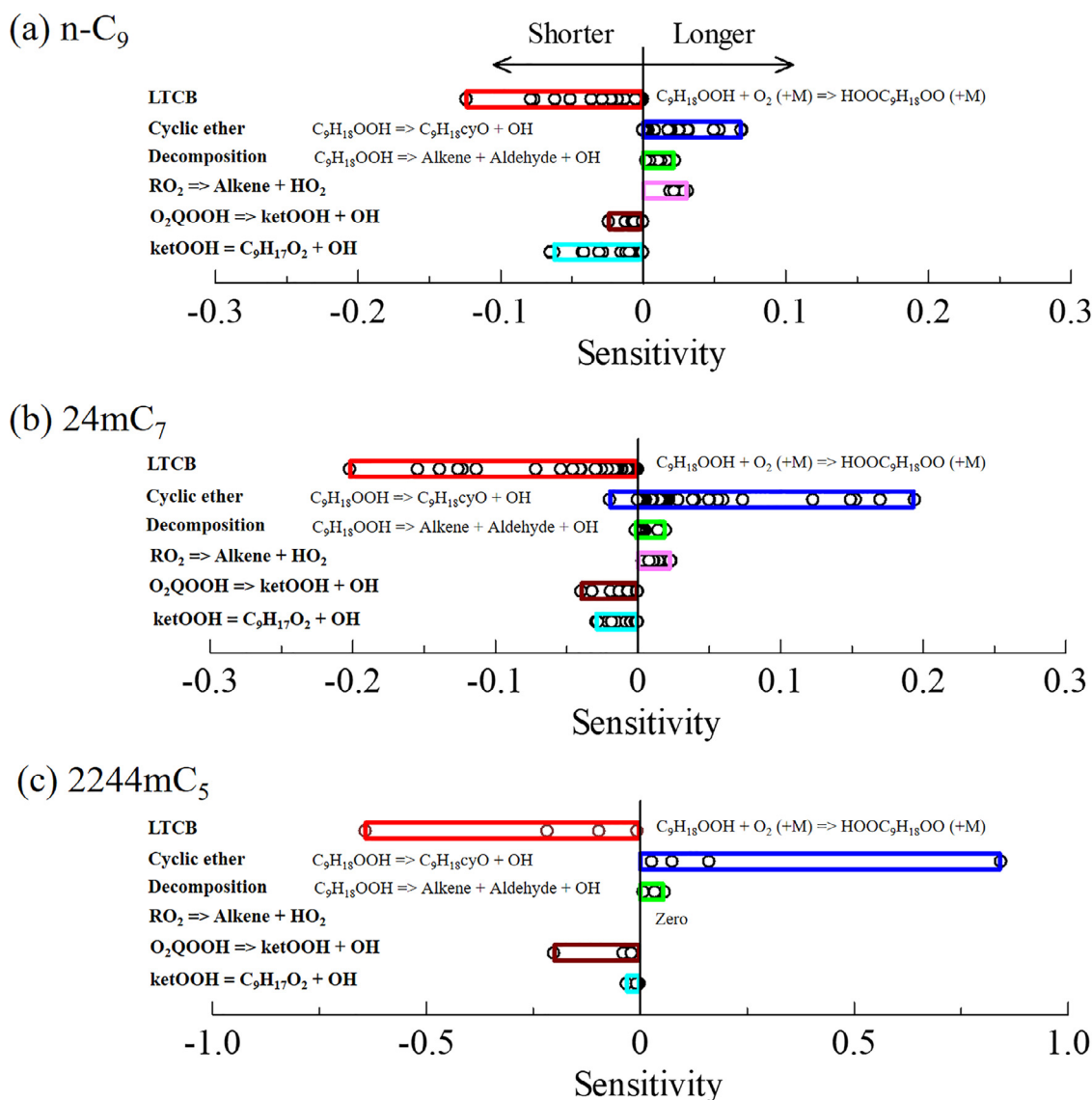


Fig. 16. Sensitivity analyses of IDT for (a) $n\text{-C}_9$, (b) 24mC_7 and (c) 2244mC_5 in O_2/Ar at $T = 750\text{ K}$, $P = 1.0\text{ MPa}$ and $\Phi = 1.0$.

for IDT even the cyclic ether formation reaction is chain propagation process because LTCB in Fig. 17(c) is prohibitably slow (see Fig. 16b, a symbol in negative position in cyclic ether formation corresponds to the reaction in Fig. 17c). Similar structural effects have already been discussed by Sarathy et al. for 2,5-dimethylhexane [23] and 2,7-dimethyloctane. However, the degree of structural inhibition effects seems to be more evident for 2,4-dimethylheptane because tertiary carbon sites are adjoined in 24mC_7 , and thus, the six-member ring isomerization is more strongly inhibited.

As for 2244mC_5 , reactions of LTCB and cyclic ether formation show considerably large absolute values of sensitivity. Reactions that proceed through two QOOH isomers are illustrated in Fig. 17(d) and (e). From Fig. 17(d) and (e), it can be seen that 2244mC_5 has only one secondary site, and therefore, the initial H-atom abstraction or the following six-member isomerization must involve H-atoms in the primary site. For this reason, the rate of low-temperature oxidation is smaller than $n\text{-C}_9$ and 24mC_7 .

5.2. Difference of laminar flame speed

Evident difference can also be recognized for laminar flame speeds of $n\text{-C}_9$ and 2244mC_5 . The effect of fuel branching on the

flame speed has been thoroughly investigated by Ji et al. [24] for C_8H_{18} isomers, $n\text{-C}_8\text{H}_{18}$, 2-methylheptane, 3-methylheptane, 2,5-dimethylhexane, and 2,2,4-trimethylpentane. They found that the n -octane/air flame exhibits the highest S_L , followed by monomethylated heptane and di-methylated hexane, and then trimethylated pentane. Increase of methyl branches resulted in a decrease of S_L . As a result of detailed analyses, it was concluded that the highest S_L of $n\text{-C}_8$ among examined isomers is mainly responsible for the vinyl radical (C_2H_3) formation formed through ethylene due to its long straight chain structure. C_2H_3 promotes flame propagation by generating active species of C_2H_2 and H, and by "second chain branching" reactions, $\text{C}_2\text{H}_3 + \text{O}_2 \rightleftharpoons \text{CH}_2\text{CHO} + \text{O}$. On the other hand, in highly branched 2,2,4-trimethylpentane / air flame, low S_L is mainly attributed to aC_3H_5 and iC_4H_8 formed by direct breaking C–C bonds, which effectively remove H radicals.

In this study, we carry out the sensitivity analysis and the reaction path analysis for $n\text{-C}_9$ and 2244mC_5 to clarify the controlling mechanism of their laminar flame speeds. Only the results of analyses for 2244mC_5 are presented upon here, since the controlling mechanisms of $n\text{-C}_9$ are similar to those of $n\text{-C}_8$ [24]. Fig. 18 shows the result of flame speed sensitivity analysis for 2244mC_5 / air flame of $\Phi = 1.05$. Reactions with positive sensitivity act to

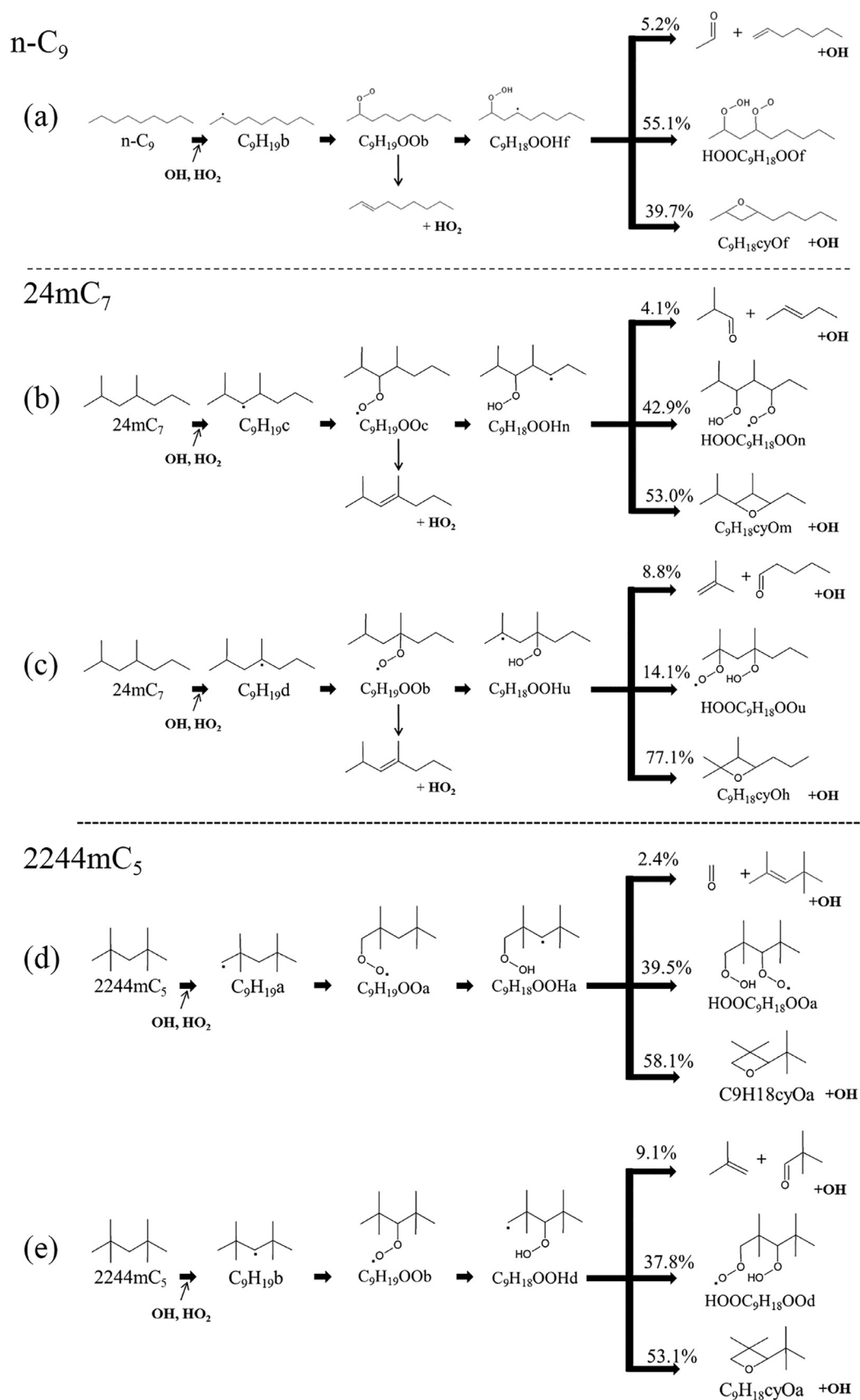


Fig. 17. Typical oxidation path of the low-temperature oxidation for (a) n-C₉, (b) 24mC₇ and (c) 2244mC₅ in O₂/Ar at $T = 750$ K, $P = 1.0$ MPa and $\Phi = 1.0$.

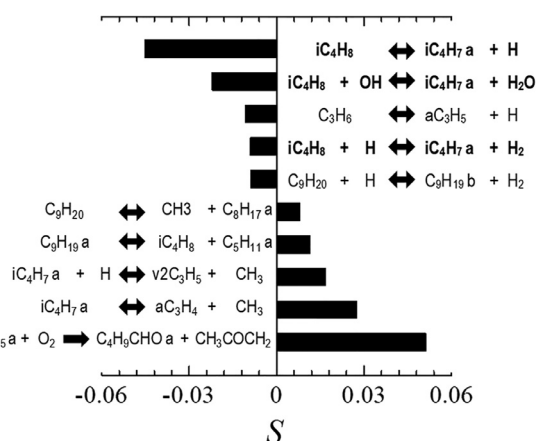


Fig. 18. Flame speed sensitivity for 2244mC₅/air flame for $T = 373$ K, $P = 0.1$ MPa and $\Phi = 1.0$, where the abscissa S represents “Sensitivity”. Note, $O_2 + H \rightleftharpoons OH + O$ ($S = 0.297$), $CO + OH \rightleftharpoons CO_2 + H$ ($S = 0.108$) and $HCO + M \rightleftharpoons H + CO + M$ ($S = 0.103$) are not shown here due to their large sensitivity.

accelerate flame speed by increasing their reaction rate, and vice versa. Note that $O_2 + H \rightleftharpoons OH + O$ ($S = 0.297$), $CO + OH \rightleftharpoons CO_2 + H$ ($S = 0.108$) and $HCO + M \rightleftharpoons H + CO + M$ ($S = 0.103$) are not shown in Fig. 18 due to their excessively large (positive) sensitivity.

As shown in Fig. 18, it is found that iC_4H_8 (isobutene) reaction has the smallest negative sensitivity for laminar flame speed among all reactions,



To clarify the controlling mechanism of S_L , reaction path analyses are made at the peak position of the iC_4H_8 fraction; it is found that iC_4H_8 is mainly formed by breaking C–C bonds of $C_9H_{19}a$ between the secondary and quaternary carbons,



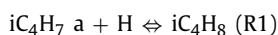
and generated $iC_5H_{11}a$ also forms iC_4H_8 by,



iC_4H_8 is also found to be formed through tC_4H_9 , which is generated by C–C bond breaking of the original fuel molecule, 2244mC₅



Generated iC_4H_8 effectively removes H and OH radicals to form iC_4H_7a , and furthermore, iC_4H_7a removes another H radical to form iC_4H_8 ,



The above radical removal loop by iC_4H_8 and iC_4H_7a is illustrated in Fig. 19. As seen, iC_4H_8 converts H or OH radicals to H_2 or H_2O on the way to form iC_4H_7a , then, H radicals are further removed when iC_4H_7a returns to iC_4H_8 . Ji et al. [24] reported that the $iC_4H_8 \rightleftharpoons iC_4H_7a$ loop as well as the $C_3H_6 \rightleftharpoons C_3H_5a$ loop are the main controlling mechanisms that slow S_L of i -C₈. However,

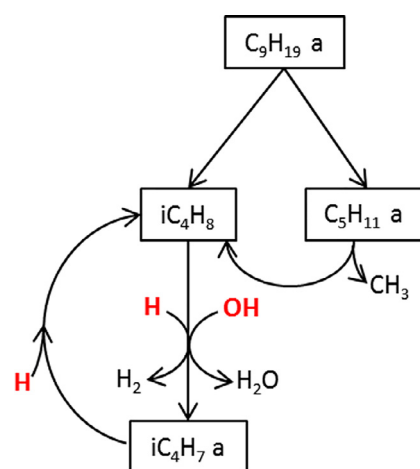


Fig. 19. Formation and looping reaction pathway of iC_4H_8 in 2244mC₅/air flame.

the effects of the $iC_4H_8 \rightleftharpoons iC_4H_7a$ loop far exceed that of the $C_3H_6 \rightleftharpoons C_3H_5a$ loop, which should serve as a radical removal loop for 2244mC₅/air flame owing to its molecular structure.

Here, it should be noted that the above analyses are made with the modified KUCRS which can reasonably reproduce S_L for n -C₉ and 2244mC₅. However, numerical IDTs with KUCRS in the high temperature region ($1.0 > 1000/T > 0.8$) underestimate experimental IDTs for n -C₉ and 2mC₈. After the sensitivity analysis and reaction path analysis, it is found that the dominant reactions are different for S_L and IDT in the high temperature region (please see the Supplemental Material 4). Further experimental / numerical research on the ignition delay times in the high temperature region for n -C₉ is still needed.

6. Conclusions

In this study, ignition delay times and laminar burning velocities of C_9H_{20} (Nonane) isomers are experimentally and numerically investigated using a newly developed shock tube facility, a dual-chamber constant-pressure/temperature spherical expanding flame rig, and numerical simulations with KUCRS, LLNL and JetSurf. It is found that experimental IDTs increase with the number of methyl branches, especially in low-temperature and negative temperature coefficient (NTC) regions. It is also found that the measured values of S_L for highly branched 2244mC₅ are smaller than those of n -C₉ at all values of Φ studied. Experimental IDT and S_L are successfully reproduced by the 0-D homogeneous reactor model and 1-D flame speed model with KUCRS.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2020.12.043.

References

- [1] H.J. Curran, Rate constant estimation for C_1 to C_4 alkyl and alkoxy radical decomposition, *Int. J. Chem. Kinet.* 38 (2006) 250–275.
- [2] M. Mehl, G. Vanhove, W.J. Pitz, E. Ranzi, Oxidation and combustion of the n-hexene isomers: a wide range kinetic modeling study, *Combust. Flame* 155 (2008) 756–772.
- [3] M. Mehl, W.J. Pitz, C.K. Westbrook, K. Yasunaga, C. Conroy, H.J. Curran, Autoignition behavior of unsaturated hydrocarbons in the low and high temperature regions, *Proc. Combust. Inst.* 33 (2011) 201–208.
- [4] A. Miyoshi, Molecular size dependent fall off rate constants for the recombination reactions of alkyl radicals with O_2 and implications for simplified kinetics of alkylperoxy radicals, *Int. J. Chem. Kinet.* 44 (2012) 59–74.
- [5] A. Miyoshi, Systematic computational study on the unimolecular reactions of alkylperoxy (RO_2), hydroperoxyalkyl ($QOOH$), and hydroperoxyalkylperoxy (O_2QOOH) radicals, *J. Phys. Chem. A* 115 (2011) 3301–3325.
- [6] D.F. Davidson, B.M. Gauthier, R.K. Hanson, Shock tube ignition measurements of iso-octane/air and toluene/air at high pressures, *Proc. Combust. Inst.* 30 (2005) 1175–1182.
- [7] Y. Sakai, A. Miyoshi, M. Koshi, W.J. Pitz, A kinetic modeling study on the oxidation of primary reference fuel–toluene mixtures including crossreactions between aromatics and aliphatics, *Proc. Combust. Inst.* 32 (2009) 411–418.
- [8] D. Darcy, C.J. Tobin, K. Yasunaga, J.M. Simmie, J. Wurmel, W.K. Metcalfe, T. Niass, S.S. Ahmed, C.K. Westbrook, H.J. Curran, A high pressure shock tube study of n-propylbenzene oxidation and its comparison with n-butylbenzene, *Combust. Flame* 159 (2012) 2219–2232.
- [9] M.K. Altarawneh, B.Z. Dlugogorski, E.M. Kennedy, J.C. Mackie, Rate constants for reactions of ethylbenzene with hydroperoxy radical, *Combust. Flame* 160 (2013) 9–16.
- [10] P.A. Glaude, V. Conraud, R. Fournet, F. Battin-Leclerc, G.M. Come, G. Scacchi, P. Dagaut, M. Cathonnet, Modeling the oxidation of mixtures of primary reference automobile fuels, *Energy Fuel* 16 (2002) 1186–1195.
- [11] K. Fieweger, R. Blumenthal, G. Adomeit, Shocktube investigation on the self-ignition of hydrocarbon–air mixtures at high pressures, *Proc. Combust. Inst.* 25 (1994) 1579–1585.
- [12] D.F. Davidson, M.A. Oehlschlaeger, J.T. Herbon, R.K. Hanson, Shock tube measurements of iso-octane ignition times and OH concentration time histories, *Proc. Combust. Inst.* 29 (2002) 1295–1301.
- [13] H.S. Shen, J. Vanderover, M.A. Oehlschlaeger, A shock tube study of iso-octane ignition at elevated pressures: the influence of diluent gases, *Combust. Flame* 155 (2008) 739–755.
- [14] M. Mehl, W.J. Pitz, C.K. Westbrook, H.J. Curran, Kinetic modeling of gasoline surrogate components and mixtures under engine conditions, *Proc. Combust. Inst.* 33 (2011) 193–200.
- [15] A. Miyoshi, Y. Sakai, Construction of a detailed kinetic model for gasoline surrogate mixture, *Trans. Soc. Autom. Eng. Jpn.* 48 (2017) 1021–1026 [The kinetic mechanism is available on <http://www.akrmys.com/kinm/>].
- [16] S. Dooley, S.H. Won, M. Chaos, J. Heyne, Y. Ju, F.L. Dryer, K. Kumar, C. Sung, H. Wang, M.A. Oehlschlaeger, R.J. Santoro, T.A. Litzinger, A jet fuel surrogate formulated by real fuel properties, *Combust. Flame* 157 (2010) 2333–2339.
- [17] M. Colket, J. Heyne, M. Rumizen, M. Gupta, T. Edwards, M. Roquemore, G. An-dac, et al., Overview of the national jet fuels combustion program, *AIAA J.* 55 (2017) 1087–1104.
- [18] W. Wang, Z. Li, M.A. Oehlschlaeger, D. Healy, H.J. Curran, S.M. Sarathy, M. Mehl, W.J. Pitz, C.K. Westbrook, An experimental and modeling study of the autoignition of 3-methylheptane, *Proc. Combust. Inst.* 34 (2013) 335–343.
- [19] K. Zhang, C. Banyon, U. Burke, G. Kukkadapu, S.W. Wagnon, M. Mehl, H.J. Curran, C.K. Westbrook, W.J. Pitz, An experimental kinetic modeling study of the oxidation of hexane isomers: developing consistent reaction rate rules for alkanes, *Combust. Flame* 206 (2019) 123–137.
- [20] C.K. Westbrook, W.J. Pitz, H.C. Curran, J. Boecker, E. Kunrath, Chemical kinetic modeling of heptane isomers, *Int. J. Chem. Kinet.* 33 (2001) 868–877.
- [21] C.K. Westbrook, W.J. Pitz, H.C. Curran, J.F. Griffiths, C. Mohamed, M. Ribaucour, Detailed chemical kinetic reaction mechanisms for autoignition of isomers of heptane under rapid compression, *Proc. Combust. Inst.* 20 (2002) 1311–1318.
- [22] S. Li, A. Campos, D.F. Davidson, R.K. Hanson, Shock tube measurements of branched alkane ignition delay times, *Fuel* 118 (2014) 398–405.
- [23] S.M. Sarathy, T. Javed, F. Karsenty, A. Heufer, W. Wang, S. Park, A. Elwardany, A. Farooq, C.K. Westbrook, W.J. Pitz, M.A. Oehlschlaeger, G. Dayma, H.J. Curran, P. Dagaut, A comprehensive combustion chemistry study of 2,5-dimethyl hexane, *Combust. Flame* 161 (2014) 1444–1459.
- [24] C. Ji, S.M. Sarathy, P.S. Veloo, C.K. Westbrook, F.N. Egolfopoulos, Effects of fuel branching on the propagation of octane isomers flames, *Combust. Flame* 164 (2016) 1426–1436.
- [25] S.M. Sarathy, C.K. Westbrook, M. Mehl, W.J. Pitz, C. Togbe, P. Dagaut, H. Wang, M.A. Oehlschlaeger, U. Niemann, K. Seshadri, P.S. Veloo, C. Ji, F.N. Egolfopoulos, T. Lu, Comprehensive chemical kinetic modeling of the oxidation of 2-methylalkanes from C_7 to C_{20} , *Combust. Flame* 158 (2011) 2338–2357.
- [26] S. Yamada, D. Shimokuri, T. Yatsufusa, T. Sakima, T. Endo, Y. Nou, F. Saito, A. Miyoshi, Measurement and numerical simulation on the ignition delay times of nonane (C_9H_{20}) isomers, *J. Combust. Soc. Jpn.* 62 (2019) 64–73.
- [27] C.J. Ji, E. Dames, Y.L. Wang, H. Wang, F.N. Egolfopoulos, Propagation and extinction of premixed C_5 – C_{12} n-alkane flames, *Combust. Flame* 157 (2010) 277–287.
- [28] B. Rotavera, A.J. Vissotski, M.C. Krejci, E.L. Petersen, Laminar flame speed measurements of methyl octanoate, n-nonane, and methylcyclohexane, 51st AIAA Aerospace Sciences Meeting Including the New Horizons Forum and Aerospace Exposition, 2013 AIAA2013-1166.
- [29] K. Yong, J. He, W. Zhang, L. Xian, C. Zhang, P. Li, X. Li, Shock tube study of n-nonane/air ignition over a wide range of temperatures, *Fuel* 188 (2017) 567–574.
- [30] J. Guo, W. Peng, S. Zhang, J. Lei, J. Jing, R. Xiao, S. Tang, Comprehensive comparison of the combustion behavior for low-temperature combustion of n-Nonane, *ACS Omega* 5 (2020) 4924–4936.
- [31] R.K. Hanson, G.A. Pang, S. Chakraborty, W. Ren, S. Wang, D.F. Davidson, Constrained reaction volume approach for studying chemical kinetics behind reflected shock waves, *Combust. Flame* 160 (2013) 1550–1558.
- [32] E.R. Ritter, J.W. Bozzelli, THERM: thermodynamic property estimation for gas phase radicals and molecules, *Int. J. Chem. Kinet.* 23 (1991) 767–778.
- [33] A. Miyoshi, KUCRS software library, revision 20110107m11, available from the author. See the web: <http://akrmys.com/KUCRS/> for update information.
- [34] C.C. Liu, S.S. Shy, H.C. Chen, M.W. Peng, On interaction of centrally-ignited, outwardly-propagating premixed flames with fully-developed isotropic turbulence at elevated pressure, *Proc. Combust. Inst.* 33 (2011) 1293–1299.
- [35] C.C. Liu, S.S. Shy, M.W. Peng, C.W. Chiu, Y.C. Dong, High-pressure burning velocities measurements for centrally-ignited premixed methane/air flames interacting with intense near-isotropic turbulence at constant Reynolds numbers, *Combust. Flame* 159 (2012) 2608–2619.
- [36] S.S. Shy, C.C. Liu, J.Y. Lin, L.L. Chen, A.N. Lipatnikov, S.I. Yang, Correlations of high-pressure lean methane and syngas turbulent burning velocities: effects of turbulent Reynolds, Damköhler, and Karlovitz numbers, *Proc. Combust. Inst.* 35 (2015) 1509–1516.
- [37] A.N. Lipatnikov, S.S. Shy, W.Y. Li, Experimental assessment of various methods of determination of laminar flame speed in experiments with expanding spherical flames with positive Markstein lengths, *Combust. Flame* 162 (2015) 2840–2854.
- [38] L.J. Jiang, S.S. Shy, W.Y. Li, H.M. Huang, M.T. Nguyen, High-temperature, high-pressure burning velocities of expanding turbulent premixed flames and their comparison with Bunsen-type flames, *Combust. Flame* 172 (2016) 173–182.
- [39] M.T. Nguyen, D.W. Yu, S.S. Shy, General correlations of high pressure turbulent burning velocities with the consideration of Lewis number effect, *Proc. Combust. Inst.* 37 (2019) 2391–2398.
- [40] A.P. Kelly, C.K. Law, Nonlinear effects in the extraction of laminar flame speeds from expanding spherical flames, *Combust. Flame* 156 (2009) 1844–1851.
- [41] ANSYS Chemkin 19.0, ANSYS, Inc.
- [42] C.K. Westbrook, W.J. Pitz, O. Herbinet, H.J. Curran, E.J. Silke, A comprehensive detailed chemical kinetic reaction mechanism for combustion of n-alkane hydrocarbons from n-octane to n-hexadecane, *Combust. Flame* 156 (2009) 181–199.
- [43] H. Wang, E. Dames, B. Sirjean, D.A. Sheen, R. Tango, A. Violi, et al., JetSurF version 2.0, (2010) [<http://web.stanford.edu/group/haiwanglab/JetSurF/JetSurF2.0/index.html>].
- [44] Y. Li, C.W. Zhou, K.P. Somers, K. Zhang, H.J. Curran, The oxidation of 2-butene: a high pressure ignition delay, kinetic modeling study and reactivity comparison with iso-butene and 1-butene, *Proc. Combust. Inst.* 36 (2017) 403–411.
- [45] C.W. Zhou, Y. Li, E. O'Connor, K.P. Somers, S. Thion, C. Keese, O. Mathieu, E.L. Petersen, T.A. DeVerter, M.A. Oehlschlaeger, G. Kukkadapu, C.J. Sung, M. Alrefae, F. Khaled, A. Farooq, P. Durrenberger, P.A. Glaude, F. Battin-Leclerc, J. Santner, Y. Ju, T. Held, F.M. Haas, F.L. Dryer, H.J. Curran, A comprehensive experimental and modeling study of iso-butene oxidation, *Combust. Flame* 167 (2016) 353–379.
- [46] S.M. Burke, U. Burke, O. Mathieu, I. Osorio, C. Keese, A. Morones, E. Petersen, W. Wang, T. DeVerter, M. Oehlschlaeger, B. Rhodes, R. Hanson, D. Davidson, B. Weber, C.J. Sung, J. Santner, Y. Ju, F. Haas, F. Dryer, E. Volkov, E. Nilsson, A. Konnov, M. Alrefae, F. Khaled, A. Farooq, O. Durreberger, O.A. Glaude, G. Battin-Leclerc, An experimental and modeling study of propene oxidation. Part 2: ignition delay time and flame speed measurements, *Combust. Flame* 162 (2015) 296–314.
- [47] S.M. Burke, W.K. Metcalfe, O. Herbinet, F. Battin-Leclerc, F.M. Haas, J. Santner, F.L. Dryer, H.J. Curran, An experimental and modeling study of propene oxidation. Part 1: speciation measurements in jet-stirred and flow reactors, *Combust. Flame* 161 (2014) 2765–2784.
- [48] W.K. Metcalfe, S.M. Burke, S.S. Ahmed, H.J. Curran, A hierarchical and comparative kinetic modeling study of C_1 – C_2 hydrocarbon and oxygenated fuels, *Int. J. Chem. Kinet.* 45 (2013) 638–675.
- [49] A. Keromnes, W.K. Metcalfe, K.A. Heufer, N. Donohoe, A.K. Das, C.J. Sung, J. Herzler, C. Naumann, P. Griebel, O. Mathieu, M.C. Krejci, E.L. Petersen, W.J. Pitz, H.J. Curran, An experimental and detailed chemical kinetic modeling study of hydrogen and syngas mixture oxidation at elevated pressures, *Combust. Flame* 160 (2013) 995–1011.
- [50] U. Burke, W.K. Metcalfe, S.M. Burke, K.A. Heufer, P. Dagaut, H.J. Curran, A detailed chemical kinetic modeling, ignition delay time and jet-stirred reactor study of methanol oxidation, *Combust. Flame* 165 (2016) 125–136.
- [51] C. Ji, E. Dames, Y.L. Wang, H. Wang, F.N. Egolfopoulos, Propagation and extinction of premixed C_5 – C_{12} n-alkane flames, *Combust. Flame* 157 (2010) 277–287.
- [52] B. Rotavera, M. Krejci, A. Vissotski, E. Petersen, Laminar flame speed measurements of methyl octanoate, n-nonane, and methylcyclohexane, AIAA Aerospace Sciences Meeting Including the New Horizons Forum and Aerospace Exposition, 2013.